

Investigation of pre-crash avoidance kinematics in pedestrians of different ages through volunteer experiment and scaling methodology

Siyuan Liu, Quan Li, Huamu Sun, Qing Zhou & Bingbing Nie

To cite this article: Siyuan Liu, Quan Li, Huamu Sun, Qing Zhou & Bingbing Nie (2025) Investigation of pre-crash avoidance kinematics in pedestrians of different ages through volunteer experiment and scaling methodology, Traffic Injury Prevention, 26:3, 281-290, DOI: [10.1080/15389588.2024.2408402](https://doi.org/10.1080/15389588.2024.2408402)

To link to this article: <https://doi.org/10.1080/15389588.2024.2408402>

 View supplementary material 

 Published online: 13 Dec 2024.

 Submit your article to this journal 

 Article views: 293

 View related articles 

 View Crossmark data 



Investigation of pre-crash avoidance kinematics in pedestrians of different ages through volunteer experiment and scaling methodology

Siyan Liu , Quan Li, Huamu Sun, Qing Zhou  and Bingbing Nie 

School of Vehicle and Mobility, State Key Laboratory of Intelligent Green Vehicle and Mobility, Tsinghua University, Beijing, China

ABSTRACT

Objective: Understanding pedestrians' pre-crash avoidance kinematics is crucial for improving the identification of potential collision areas in interactions with highly automated vehicles (HAVs). Age significantly influences pedestrian avoidance velocity and the subsequent crash risks. However, current active safety systems in HAVs often overlook the influence of pedestrians' avoidance velocity and age on imminent accidents. This study analyzes how age affects pedestrian avoidance velocity and explores the incorporation of these factors in pre-crash scenarios to identify potential collision areas between pedestrians and vehicles.

Methods: Due to the infeasibility of measuring pedestrian avoidance behaviors in real-world pre-crash scenarios, we designed an indoor experimental platform replicating emergency crossroad scenarios to prompt subjects to mimic avoidance behaviors. 7 young and 7 middle-aged subjects participated in the experiment, resulting in a collection of 306 forward-avoidance, 297 backward-avoidance, and 42 normal-walking posture sequences. We developed a scaling approach integrating pedestrian kinematics and muscle physiology to establish a velocity-mapping relationship between young and middle-aged groups. Finally, we proposed an identification method for potential collision areas that considers pedestrians' age and avoidance velocity.

Results: Middle-aged subjects required more time for natural avoidance actions averaging 0.15 s for forward and 0.25 s for backward avoidance, compared to their younger counterparts. While the forward avoidance velocity of the middle-aged subjects exhibited an average decrease of 0.3 m/s compared to young subjects, their backward avoidance velocity remained nearly identical. Overall, middle-aged subjects have a larger potential collision area than young participants. Pedestrians who actively avoid vehicles have a smaller potential collision area compared to those who remain normal walking.

Conclusions: We developed an indoor simulated pre-crash scenario experiment and a scaling approach to reveal the correlation between pedestrian avoidance velocity and age. This method can be further applied to obtain the avoidance velocity of elderly pedestrians. Additionally, we validate the effect of these factors in assessing potential collision areas. The decrease in avoidance velocity highlights a larger potential collision area for middle-aged pedestrians when interacting with vehicles. Such facts and data shall be appropriately considered in developing intelligent protection systems for pedestrians.

ARTICLE HISTORY

Received 8 July 2024

Accepted 20 September 2024

KEYWORDS

Pedestrian safety; age; pre-crash behavior; collision area identification; highly automated vehicles

Introduction

As vulnerable road users (VRUs), pedestrians remain at high injury risk, accounting for 23% of global-reported road deaths (WHO, 2023). Active safety systems are commonly employed in current highly automated vehicles (HAVs) to reduce the severity of pedestrian injuries (Ye et al. 2024). Pedestrians' age and behaviors, such as changes in avoidance velocity in pre-crash scenarios, significantly influence the occurrence and severity of the accidents (Kim et al. 2008; Soni et al. 2013). Yet, such factors have seldom been considered in current safety systems in HAVs due to the lack of an accurate description of the underlying mechanisms (Coelingh et al. 2010; Edwards et al. 2015; Park et al. 2016).

Accurately quantifying the avoidance velocity of pedestrians of varying ages in pre-crash scenarios is crucial for active safety systems to identify potential collision areas. However, accident databases gather data on injuries and fatalities but do not offer quantitative information on the change in pedestrians' velocity (e.g., German In-Depth Accident Study, GIDAS). Additionally, ethical constraints make recruiting participants for real pre-crash experiments impossible, resulting in a lack of data describing the pedestrians' velocity before the collision. This lack of quantitative analysis may lead to highly automated vehicle manufacturers making inappropriate decisions while designing protection strategies for pedestrians.

CONTACT Bingbing Nie  nbb@tsinghua.edu.cn  School of Vehicle and Mobility, State Key Laboratory of Intelligent Green Vehicle and Mobility, Tsinghua University, Beijing 100084, China.

Associate editor Jason Forman oversaw the review of this article

 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/15389588.2024.2408402>.

© 2024 Taylor & Francis Group, LLC

Unlike typical pedestrian walking behaviors (Tian et al. 2022; Gerogiannis and Bode 2024), avoidance behaviors involve moving forward or backward due to the survival instinct to avoid collisions (Li et al. 2021). Schachner et al. (2020) used computer vision approaches to extract pedestrian behaviors from road cameras. It found that typical pre-crash reactions, such as leaning backward, significantly changed the pedestrian's avoidance velocity. Simulated pre-crash scenario experiments were constructed with volunteers using a non-impacting methodology. Through the experiment, Soni et al. (2013) found that the participants exhibited reactions of running, stepping back or freezing in fear when confronted with hazardous situations. Advanced virtual reality technology was used to immerse volunteers in a virtual pre-crash scenario to enhance the authenticity. Subjects' avoidance behaviors were recorded to analyze the potential collision areas between young pedestrians and vehicles (Nie et al. 2021). The above two research studies indicate that pedestrian avoidance behaviors, which are distinct from normal walking, can be measured through simulated scenario experiments that do not rely on road accident videos.

Another factor to consider is the strong correlation between age and pedestrian velocity (Knoblauch et al. 1996), as well as the severity of injuries after collision (Niebuhr et al. 2016; Oikawa et al. 2016). Pedestrians with higher age may not take velocity decrease into account when making decisions similar to the young, leading to an increased collision risk (Liu and Tung 2014). After collisions, the likelihood of a pedestrian suffering fatal injuries rises with age (Kim et al. 2008). However, above studies have not analyzed how age influences pedestrian avoidance velocity and collision risk prior to the crash. This gap may result from the absence of real data on elderly pedestrians as they are more prone to falls and injuries in experiments. In biomechanics, scaling approaches are commonly used to address the dynamic response of different-aged people. For example, Zhou et al. (1996) and Gunji et al. (2019) separately used piecewise linear regression and scaled human body models to analyze chest injuries in young, middle-aged and elderly people. These studies inspire that by exploring non-linear age relationships with research objectives (e.g., velocity or injury severity), data from the young can be mapped to the middle-aged even elderly.

To identify statistical evidence of the non-linear relationship between pedestrian avoidance velocity and age, the medical field has collected extensive age-related data on pedestrian gait kinematics and physiology. Regarding kinematic analysis, Grimmer et al. (2019) investigated the maximum walking velocity of pedestrians aged 24 to 75. Their findings revealed that elderly pedestrians exhibited a 24% reduction in velocity compared to young pedestrians. From the physiological perspective, Aoyagi and Shephard (1992) discovered that as age increases, the contraction time and half-relaxation time of lower limb muscles are prolonged, leading to a decrease in the contraction speed of these muscles. In addition, lower limb muscle strength tends to decrease with age, with reductions ranging from 12.9% to 25.3% (Pasco et al. 2020). Der and Deary (2006) established

models for age-related differences in reaction times (RTs), demonstrating that RTs decrease with age. These findings indirectly reveal the decreasing trend of pedestrian avoidance velocity with age, but the specific quantitative relationships remain unclear.

The present study aims to analyze different aged pedestrians' avoidance velocity in pre-crash scenarios. The problem was decomposed as follows: (i) Develop and validate an indoor experiment to examine the disparity in avoidance velocity between young and middle-aged pedestrians. (ii) Analyze the non-linear relationship between avoidance velocity and age among pedestrians and propose a scaling approach for avoidance velocity using data from young pedestrians as a foundation. (iii) Considering pedestrian avoidance velocity to calculate the potential collision areas between pedestrians and vehicles for highly automated vehicles and compare the impact of age on these areas.

Methods

An indoor crossroad scenario was established to replicate the boundary conditions of real-world vehicle-pedestrian pre-crash scenarios in a controlled environment (Figure A1, Phase I, supplementary file). It stimulated and recorded the avoidance velocity of young and middle-aged subjects. To overcome the potential risks (e.g., falls and dizziness) of experimental stimulus on elderly subjects, we proposed an avoidance velocity scaling approach to avoid harm to such subjects. As an illustration, we exhibited the non-linear relationship between young and middle-aged pedestrians (Figure A1, Phase II, supplementary file). Finally, pedestrian avoidance velocity was employed to identify potential collision areas in pre-crash scenarios between pedestrians and vehicles, and we performed a quantitative analysis of how age influences these areas (Figure A1, Phase III, supplementary file).

Avoidance velocity acquisition via the simulated scenario experiment

Experiment design and kinematic capture system

In real-world pedestrian-vehicle collisions, pedestrians' responses and muscle contraction capacity play a critical role in changing their behaviors. Under emergency situations such as an imminent accident, pedestrians instinctively respond either voluntarily or involuntarily, utilizing their survival instincts to avoid the collision by moving forward or backward (Figure 1a-1b). The experiment aims to stimulate pedestrians of different ages to imitate the behaviors of avoiding approaching vehicles while crossing the road, capturing their avoidance velocity on a large scale (Figure 1c). A crossroads scenario was constructed in a well-protected indoor space measuring 8 meters in length and 5 meters in width. This scenario integrated vital elements relevant to the interaction between pedestrians and approaching vehicles in real-world pre-crash scenarios. Specifically, these elements included a 2-meter hazardous zone, which was aligned with pedestrian motion direction to simulate the width of a

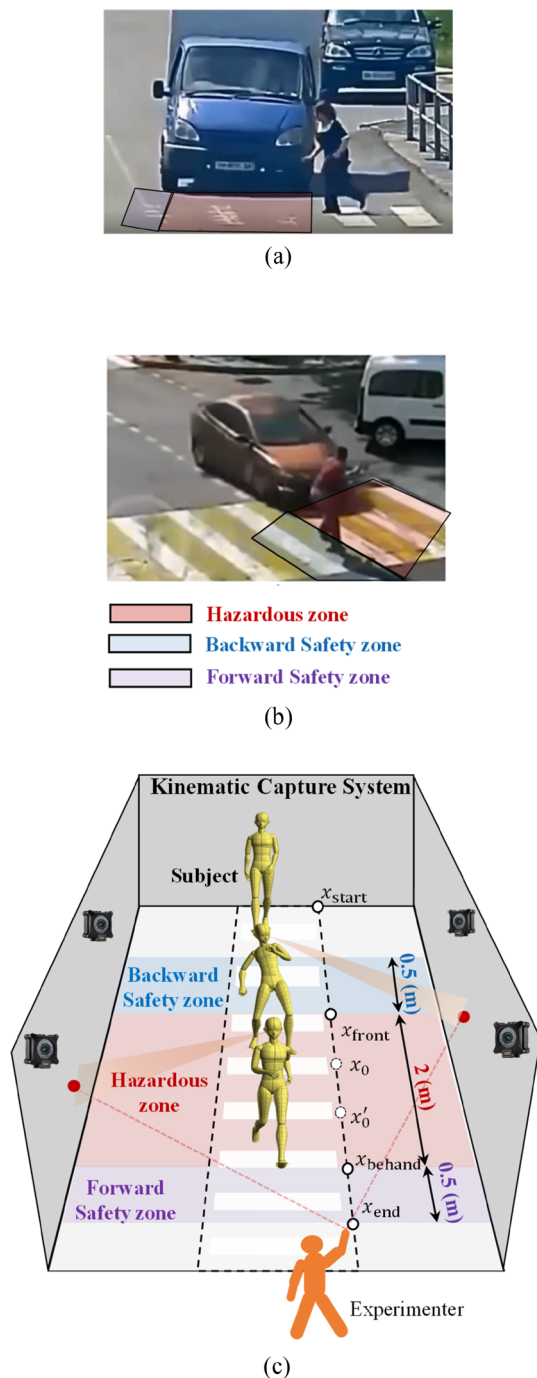


Figure 1. (a) Forward avoidance behavior in real traffic accident; (b) backward avoidance behavior in real traffic accident; (c) simulated scenario experiment (the random red light from the experimenter represents a vehicle approaching perpendicular to the pedestrian's direction; the red zone represents the collision area; the blue and purple zones represent safe areas for pedestrians after avoiding approaching vehicles).

vehicle. Furthermore, random red light stimuli given by an experimenter on one side of the wall were used to simulate the sudden appearance of a vehicle in a real pre-crash scenario (i.e., In real-world collisions, pedestrians often fail to notice approaching vehicles promptly, just as the red light suddenly appeared in the subjects' sight). After observing the red light, the subjects demonstrated avoidance behaviors by moving either forward or backward depending on where

they were positioned in relation to the hazardous zone. To assist subjects to estimate the avoidance distance, backward and forward safety zones (0.5 meters on each side of the hazardous zone) were implemented. Subsequently, subjects were instructed to enter the scenario as "pedestrians" with standardized fitted clothing, i.e., a sleeveless sports vest and knee-length shorts, specifically designed to facilitate the attachment of on-body markers (Figure A2, supplementary file).

The kinematic capture system (No. Mars 2H; Beijing Nokov Science & Technology Co., Ltd) includes 24 cameras with a sampling frequency of 60 Hz, 54 on-body markers, and a data processing software. During the experiment, the cameras were positioned on fixed trestles at the room's edges to record subjects' kinematic information. Additionally, 54 on-body markers were strategically placed to allow the camera to capture the subjects' movements throughout the entire event. The pedestrian velocity was then extracted from the data collected at the center of the pelvis.

Volunteer information and test procedure

Subjects and ethics statement. Subjects from Tsinghua University (population about 100,000) were recruited through the Shuimu Gang (i.e., the largest online volunteer recruitment platform on campus). The inclusion criteria specified normal (or corrected to normal) vision, hearing, and walking gait without disabilities or heart disease. This study restricted admission to male individuals between 18 and 55 years. A total of 14 subjects (Table A1-A2, supplementary file), comprising 7 young adults (age range 21.2 ± 1.8 years; BMI range 22.7 ± 1.6) and 7 middle-aged adults (age range 44.3 ± 5.7 years; BMI range 25.7 ± 3.4), were involved in the experiment. The subjects were instructed that the purpose of the experiment was to imitate standard avoidance behaviors (backward or forward) to capture their avoidance velocity upon observing the stimulus. They could withdraw from the experiment if they experienced any discomfort. The experiment was approved by the Institutional Review Board (IRB) of Tsinghua University, and all experiments complied with relevant guidelines.

Test procedure. The experimenter initiated the kinematic capture system at the beginning of experiment to record the subjects' kinematic information. For each participant, the experimental process was segmented into three stages:

Stage 1 (Imitation learning): Each subject was instructed to watch two videos demonstrating the standard avoidance behaviors of pedestrians at the pre-crash scenario, which were obtained from our previous virtual reality study (Nie et al. 2021). The subjects then spent 30 minutes observing and imitating both backward and forward avoidance behaviors. Following this, they were asked to position themselves at the starting point of the scenario and await start signal.

Stage 2 (Normal walking): Upon observing a blinker signal given by the experimenter, subjects were instructed to walk straight ahead in a normal walking gait. They were asked to cautiously observe the walls on both sides of their forward

direction, imitating the behavior of observing the traffic environment as if they were navigating through a real crossroad. The normal walking stage was conducted three times for each subject. During this process, subjects were required to promptly stop and remain stationary upon sighting a green light signal on one side of the wall, ensuring they were carefully observing the surrounding environment.

Stage 3 (Imitation avoidance): In contrast to stage 2, as subjects commenced walking, the experimenter randomly presented a red light signal on either side of the wall. Upon spotting the red light signal, subjects were required to determine whether to avoid forward or backward rapidly based on their position relative to the hazardous zone. Subjects were instructed to move into the safety zones to ensure that they replicated natural avoidance behaviors rather than engaging in vigorous movements. For each subject, the experiment would be repeated 25 times for both forward and backward avoidance, and the kinematics data of subjects during the three stages were captured as displacement data using the motion capture system.

Data and validation

A kinematic database was established to represent the motion ability of two age groups by collecting 603 pieces of forward and backward avoidance data and 42 pieces of normal walking data. For each subject, average 43 pieces of avoidance data and 3 pieces of normal walking data were gathered, and avoidance velocity corridors for each age group were calculated. As a result, the database effectively describes both avoidance velocity uncertainty for individual subjects and the avoidance corridors for each age group in the experiment. In order to ensure data validity, we meticulously selected critical elements from real pre-crash scenarios, thereby enhancing the fidelity of the experiment. These elements included abrupt hazardous stimuli, a fixed hazardous zone, random perception of pedestrians, and autonomous avoidance decision-making in avoiding the hazardous vehicle. To verify the accuracy of our experiment, we assumed that pedestrians in a specific age group had a normal distribution of avoidance velocity. Then, we compare the avoidance velocities obtained in our study and the velocity corridor of young subjects in a similar study with virtual reality (VR) scenarios to assess the consistency of our data. The experiment was deemed to have a high level of similarity if the avoidance velocities fell within the 2σ principle at any point within the corridor.

Scaling approach inspired by gait analysis

Unlike the young and middle-aged subjects, the experiment did not include elderly subjects (over 55 years old) due to the hazardous stimuli risk. Therefore, based on the experimental data from the young subjects, it is imperative to develop a scaling approach for avoidance velocity that accounts for age differences among pedestrians. This section demonstrates and validates the approach by illustrating the scaled avoidance velocities between young and middle-aged subjects.

Gait analysis and mathematical representation

Gait analysis could describe pedestrian walking motion (Harris and Wertsch 1994; Iosa et al. 2013; Zou et al. 2022). To quantify subjects' walking velocity, we simplify the gait into deceleration, transition, and acceleration phases. According to the above definition, a subject's pelvis velocity can be described as a gait equation:

$$v_{\text{pelvis}}(t) = \begin{cases} C + a_1 t, & 0 \leq t \leq t_1 \\ C + a_1 t_1 + a_2 (t - t_1), & t_1 \leq t \leq t_2 \end{cases} \quad (1)$$

where v_{pelvis} is the pelvis velocity during one gait, C is the initial peak velocity, a_1 is the average deceleration, a_2 is the average acceleration, t_1 is the transitional time, and t_2 is the ending time of the gait.

The pedestrian kinematics during the experiment were observed to occur in two stages: normal walking and avoidance. The pelvis velocity during each avoidance behavior exhibited two distinct gaits (e.g., Gait 1 and Gait 2 in Figure A3(d), supplementary file). Notably, the parameters (C , a_1 , a_2 , t_1) were found to be potentially influenced by kinematic and muscle physiology, including the average walking ability, hip flexor contraction force, hip abductor contraction force, ankle joint flexor muscle contraction and half-relaxation time (Aoyagi and Shephard 1992; Grimmer et al. 2019; Pasco et al. 2020). These findings indicate that pedestrian walking ability highly depends on age and established a bridge to scale avoidance velocity among pedestrians of different ages.

Scaling approach between young and middle-aged pedestrians

The parameters within gait kinematics equations enable the quantification of the variance in avoidance velocity between pedestrians of different ages. In order to illustrate the universality of our proposed model, this section will demonstrate the effectiveness of the scaling approach by using Equation (1) to scale 14 average avoidance velocities from young subjects to middle-aged subjects (i.e., 7 forward and 7 backward avoidance velocities in the experiment).

Scaling factors. The main difference in pedestrian avoidance velocity related to age lies in the differences in muscle response and motion capabilities. Pedestrians show distinct muscle responses from normal walking to avoidance gait during the transition phase. Specifically, the transition phase occurs in the middle of the normal walking gait (Figure A3(b), supplementary file) but appears earlier in the forward avoidance gait and later in the backward avoidance gait. Meanwhile, middle-aged subjects generally exhibit a delayed transition phase compared to young subjects. This difference observed between the two age groups can be attributed to lower limb muscle physiology variations, including the impact of muscle contraction and relaxation time on transition phases during avoidance gait (Figure A3(c), supplementary file). Moreover, the motion discrepancies among pedestrians could be reflected during normal walking

(Figure A3(a), supplementary file), including the initial peak velocity C , average deceleration a_p , and average acceleration a_2 . To quantitatively distinguish between the two groups, we calculate the average normal walking gait velocity of young and middle-aged pedestrians using data from our experiment. Equation (1) fits the average gait velocities, from which two sets of kinematic parameters are derived to represent pedestrians' motion abilities. Furthermore, through physiology and kinematic analysis, the scaling factors are derived from the separate division of the two sets of parameters (Equations (A3-A6)), capturing the disparity between the two age groups. Consequently, these scaling factors systematically depict the muscle response and motion variation in the avoidance gait of young and middle-aged pedestrians.

Scaling process. The avoidance duration of each young subject within the experiment was standardized to the average avoidance gait of the young group. This process yielded two avoidance gait equations, denoted as v_{Gait_y} and v_{Gait_2y} (Equations (A7-A8)), which described the avoidance velocity of young subjects. Subsequently, the kinematic parameters of each avoidance gait were multiplied by scaling factors to derive the avoidance gait equations for middle-aged subjects, expressed as v_{Gait_1m} and v_{Gait_2m} (Equations (A9-A10)). Through this process (Figure A3(d), supplementary file), the proposed model effectively scaled the avoidance velocity of young to middle-aged pedestrians, accounting for both kinematic and physiological differences between the two age groups.

Model validation

The accuracy of the scaling results was validated through the simulated scenario experiment, specifically by comparing the avoidance velocity corridor of middle-aged pedestrians and scaled velocity. The experimental data demonstrated that the avoidance velocity corridor of middle-aged subjects exhibits a normal distribution characteristic. In order to make a quantitative comparison, the duration of the acceleration phase of the second scaled avoidance gait (Equation (A10)) was extrapolated to ensure consistency between the scaled gait duration and middle-aged subjects' avoidance duration in our experiment (i.e., forward duration: 1.01s; backward duration: 1.64s). As such, confirmation of the model's validity hinges on the adherence of the scaled velocity to the 2σ normal distribution principle of middle-aged pedestrians' avoidance velocity corridor.

Age influence on potential collision areas identification

To quantitatively analyze the influence of age on pedestrian safety, we assessed potential collision areas between pedestrians of different ages and vehicles. We specifically focused on two common scenarios: (1) instances in which pedestrians did not perceive approaching vehicles and proceeded to cross the intersection at a normal walking gait, and (2) cases in which pedestrians did notice the presence of a hazardous

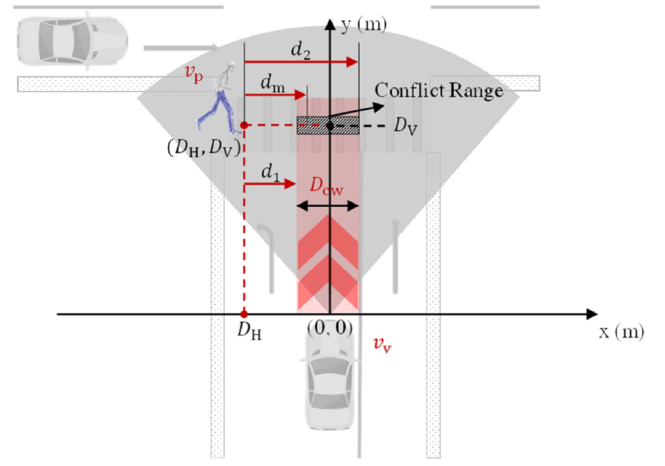


Figure 2. The typical pre-crash scenario.

vehicle and chose to either move forward or backward to avoid it. The determination of the potential collision area between pedestrians and vehicles involved calculating the relative distance relationship in both the horizontal (x) and longitudinal (y) directions (Figure 2). It should be noted that at any given time t , the pedestrian's velocity can be represented as a piecewise function.

$$v_p(t) = \begin{cases} v_N(t), c=0; 0 \leq t \leq t_N \\ v_{\text{Gait}_1}(t), c=1; t_N \leq t \leq t_{G_1} \\ v_{\text{Gait}_2}(t), c=1; t_{G_1} \leq t \leq t_{G_2} \\ v_a(t), c=1; t_{G_2} \leq t \end{cases} \quad (2)$$

where $c=0/1$ represents whether the pedestrian noticed the approaching vehicle, v_p is the pedestrian velocity, v_N is the normal walking gait velocity, v_{Gait_1} is the first avoidance gait velocity, v_{Gait_2} is the second avoidance gait velocity, v_a is the velocity after avoidance, t_N is the time when the pedestrian noticed the vehicle, t_{G_1} is the ending time of the first avoidance gait, t_{G_2} is the ending time of the second avoidance gait.

The collision happens when both the pedestrian and the vehicle reach the conflict range at the same time (Figure 2).

$$D_H - \frac{1}{2}D_{cw} \leq \int_0^{t_v} v_p(t) dt \leq D_H + \frac{1}{2}D_{cw} \quad (3)$$

where D_H is the horizontal distance between the pedestrian and the vehicle's center axis, D_{cw} is the vehicle width, t_v is the time when the vehicle arrives at the conflict range.

Thus, the potential collision areas between pedestrians of different ages and vehicles can be identified based on pedestrian mobility.

$$\int_0^{t_v} v_{p_y}(t) dt - \frac{1}{2}D_{cw} \leq D_H \leq \int_0^{t_v} v_{p_y}(t) dt + \frac{1}{2}D_{cw} \quad (4)$$

$$\int_0^{t_v} v_{p_m}(t) dt - \frac{1}{2}D_{cw} \leq D_H \leq \int_0^{t_v} v_{p_m}(t) dt + \frac{1}{2}D_{cw} \quad (5)$$

Where v_{py} is the velocity of the young pedestrian, v_{pm} is the velocity of middle-aged the pedestrian.

Results

Verification and result of the simulated scenario experiment

In order to validate the experiment, a comparison of avoidance data between young subjects in our study and a similar virtual reality-based study (Nie et al. 2021) was conducted. In their study, the researchers compiled a library of 100 real-world accident videos sourced from public online platforms and in-vehicle cameras. Pedestrian avoidance behaviors observed in volunteer experiments with those recorded in accident video sequences were qualitatively compared. Their reactions were categorized into three types per the kinematics characteristics: no response, horizontal avoidance, and vertical avoidance, consistent with those in the accidents videos (Li et al. 2021). Among them, pedestrian reactions such as no response and vertical avoidance frequently fail to prevent collisions with vehicles. Consequently, this study only focuses on two effective avoidance behaviors, i.e., forward and backward avoidance. To ensure comparability, the avoidance durations of the subjects were standardized to match a uniform avoidance gait, and subsequently, average avoidance velocity curves for each subject were calculated. The results revealed that the average avoidance velocity curves of young subjects in the experiment were, on average, within the avoidance corridor in the VR-based study (Figure A4, supplementary file), with an average 96.23% accuracy. More interestingly, the trends of these velocity curves are consistently aligned. These finding suggests that the proposed experiment can effectively capture critical elements in the pre-crash scenario and elicit similar avoidance behaviors from the subjects.

The experiment effectively yielded a total of 156 forward and 155 backward avoidance velocity curves for young subjects and 150 forward and 142 backward avoidance velocity curves for middle-aged subjects (Table A1-A2, supplementary file). These data were then subjected to statistical analysis to derive the avoidance velocity corridors for both the young and middle-aged pedestrians. The average forward avoidance velocity curve for young subjects exhibited a two-gait acceleration, with the first gait lasting an average of 0.54s (Gait₁^{YF} from time 0 to time t_1^{YF} , Figure 3a), followed by a second gait lasting 0.32s (Gait₂^{YF} from time t_1^{YF} to time t_2^{YF} , Figure 3a). Correspondingly, the average backward avoidance velocity curves for the same age group showed a two-gait deceleration, with the first gait averaging 0.99s (Gait₁^{YB} from time 0 to time t_1^{YB} , Figure 3b), and the second gait lasting 0.42s on average (Gait₂^{YB} from time t_1^{YB} to time t_2^{YB} , Figure 3b). As for the middle-aged, the first forward avoidance gait averaging 0.64s and the subsequent gait lasting about 0.37s (Gait₁^{MF} from time 0 to time t_1^{MF} and Gait₂^{MF} from time t_1^{MF} to time t_2^{MF} , Figure 3a). Similarly, their backward avoidance behaviors also display a two-gait motion, with the first gait averaging 1.02s, followed by a second gait

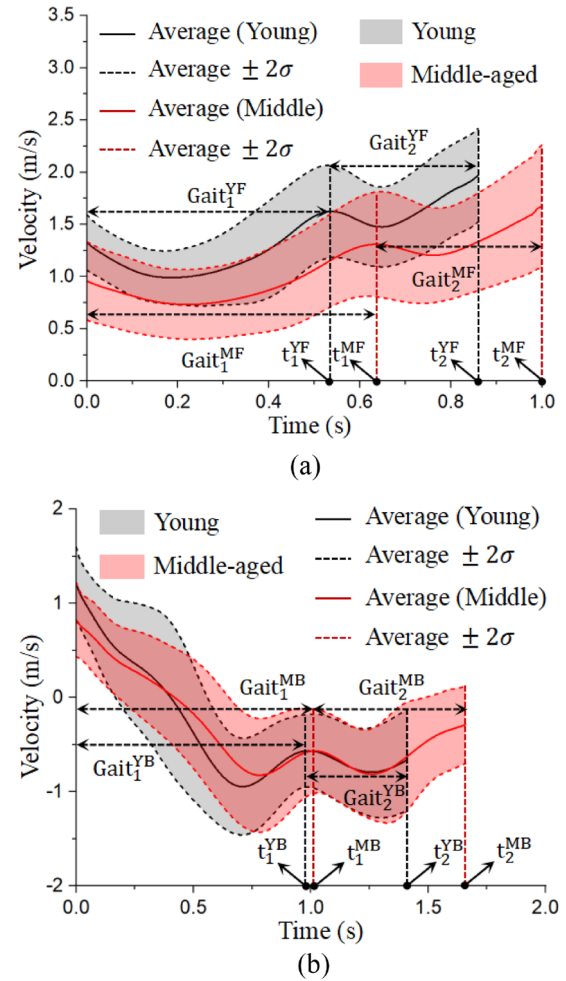


Figure 3. (a) Forward avoidance velocity corridor for young and Middle-aged subjects in the simulated scenario experiment; (b) backward avoidance velocity corridor for young and Middle-aged subjects in the simulated scenario experiment.

lasting around 0.64s on average (Gait₁^{MB} from time 0 to time t_1^{MB} and Gait₂^{MB} from time t_1^{MB} to time t_2^{MB} , Figure 3b). This indicates that middle-aged pedestrians require more time to avoid an approaching vehicle.

Verification and results of the scaling approach

The effectiveness of the experiment in capturing pedestrian avoidance velocity is demonstrated in the above section. This enables the verification of the scaling approach through a quantitative comparison of experimental data and scaled data. The 14 scaled curves all fell within the middle-aged subjects' avoidance corridors (Figure A5, supplementary file), suggesting that this approach could effectively scale the avoidance velocity of approach subjects to other age groups within 2σ principle of normal distribution. This indicates that, by analyzing the kinematics and physiology of normal walking gait, a non-linear scaled relationship between walking capability in young and middle-aged subjects can be quantitatively determined. The scaling approach derives the avoidance velocity of middle-aged subjects by utilizing the non-linear mapping relationship based on normal walking

gait velocity for both young and middle-aged subjects, as well as avoidance gait velocity for young subjects. The scaled forward and backward avoidance durations of middle-aged subjects were found to be 0.96 and 1.55 s, respectively. Compared with the durations observed in middle-aged subjects during our experiment, the scaled forward and backward avoidance durations represent 94.1% and 93.4% of the experimental values. This methodology addresses the challenge of involving subjects of high age in the stimulus experiment (i.e., utilizing the scaling approach to obtain the avoidance velocity of the elderly, as they are unable to directly measure avoidance velocity due to the risk of falls and injuries in the experiment).

Potential collision areas between pedestrians of different ages and vehicles

The calculation of potential collision areas between pedestrians and vehicles can be more accurate by considering their motion ability and age, which helps identify potential collision risks. Based on data collected from the experiment, Equations (4-5) can be used to calculate these potential collision areas for young and middle-aged subjects. When

pedestrians are unaware of an approaching vehicle and continue walking normally, the potential collision area can be represented as a quadrilateral (e.g., $I_{Y,Walking}^*$ and $II_{M,Walking}^*$ shown in Figure 4a). Conversely, pedestrians who notice the approaching vehicle and take avoidance action experience a significant reduction in their collision areas (e.g., $I_{Y,Avoid}$ and $II_{M,Avoid}$ shown in Figure 4a). Furthermore, it is observed that young subjects have smaller potential collision areas compared to middle-aged subjects (e.g., $I_{Y,Avoid}$ and $II_{M,Avoid}$ or $I_{Y,Walking}^*$ and $II_{M,Walking}^*$ shown in Figure 4a). More interestingly, as the velocity of the vehicle increases, these areas will expand, regardless of whether pedestrians avoid or walk normally. For subjects executing avoidance behaviors, as the vehicle's velocity increased from 40 km/h to 80 km/h, the furthest point of the potential collision area for the young increased by 12.78 m, while increasing 14.78 m for the middle-aged (Figure 4a-4c). These findings suggest that middle-aged subjects face a higher risk in pre-crash scenarios and that the safety protection system should implement a more flexible braking strategy (i.e., light or full braking) for pedestrians engaged in avoidance behaviors.

Discussion

Potential collision areas impact the occurrence of the crash and pedestrian injury risk. We proposed an indoor simulated scenario experiment and a scaling approach to quantitatively describe such areas across various pedestrians of different ages. Compared with existing experiments (Soni et al. 2013; Nie et al. 2021), this method enables the large-scale collection of data on pedestrian avoidance velocities across a diverse age spectrum. Furthermore, the impact of age on pedestrians' avoidance capability will eventually be reflected in potential collision areas in the pre-crash scenario.

Effectiveness and performance of the simulated scenario experiment

The present study created a safe, simulated scenario experiment in an indoor setting by extracting the key elements of the pre-crash scenarios. To achieve this, we directed subjects to mimic the most common backward and forward avoidance behaviors observed in pre-crash accidents, aligning with findings from our previous studies. The experiment involved seven young subjects, and we found that their average avoidance velocity curves align statistically with the findings of the previous VR-based study. This suggests that simplified pre-crash scenarios can effectively elicit avoidance behaviors in subjects that closely resemble those observed in real accidents. While our experiment may have sacrificed realism in pre-crash scenarios, it successfully extended the age constraint and non-repeatable challenges encountered in the prior research (Nie et al. 2021). Notably, research on muscle strength and ground contact posture (Kerrigan et al. 1998; Pasco et al. 2020) indirectly suggests expected differences in avoidance velocity between males and females. However, the VR-based study did not measure avoidance velocity for females. Consequently, only male subjects were included in this study.

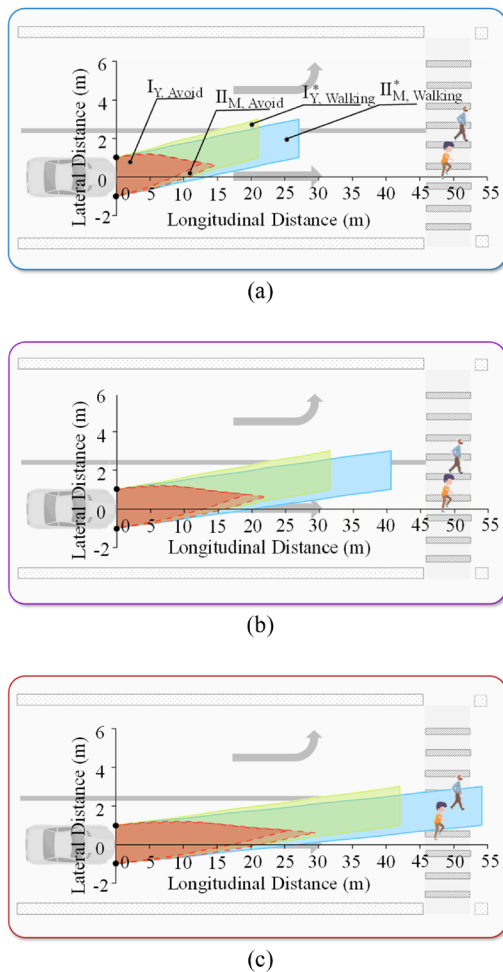


Figure 4. (a) Potential collision areas under 40 km/h ($I_{Y,Avoid}$ and $II_{M,Avoid}$ for young and middle-aged pedestrians with avoidance behavior; $I_{Y,Walking}^*$ and $II_{M,Walking}^*$ for young and Middle-aged pedestrians with normal walking); (b) potential collision areas under 60 km/h; (c) potential collision areas under 80 km/h.

We also compared the average avoidance curves between young and middle-aged subjects, revealing that middle-aged subjects need more time to avoid oncoming vehicles. This indicates that middle-aged pedestrians are at a higher risk of colliding with dangerous vehicles compared to young pedestrians when the time to collision and relative position between pedestrians and vehicles are the same. Notably, middle-aged subjects exhibited a significant decrease in average forward avoidance velocity while maintaining nearly consistent velocity for average backward avoidance. This observation suggests a potential directional shift during backward avoidance, possibly constraining subjects' motion ability. Our statistical analysis of these curves allowed us to calculate the forward and backward velocity corridors for young and middle-aged pedestrians, providing priori information for analyzing potential collision areas.

Effectiveness and application of the scaling approach

This study proposes an approach to establish a scaling relationship for avoidance velocity between young and other age groups. As an example, a total of 14 avoidance velocity curves from young subjects were scaled to the middle-aged group. Compared to the experiment data, the scaled avoidance duration shows slight differences, which is due to the random nature of the subjects' avoidance ability. To make a valid comparison, we extrapolated the acceleration phase duration of the second avoidance gait of middle-aged subjects after scaling (Equation (A10)), ensuring that the scaled velocities matched the avoidance duration in our experiment. Results showed that all 14 scaled curves fell within the avoidance velocity corridors of the middle-aged subjects (Figure A5, supplementary file). Consequently, it is evident that the proposed approach can quantify and scale pedestrian velocity under standard avoidance behaviors within the 2σ principle of normal distribution.

Our proposed model, which utilizes kinematic parameter scaling, has successfully addressed the age-related constraints typically encountered in stimulus experiments. It produced accurate results for the avoidance velocity of middle-aged pedestrians, with all 14 scaled curves mentioned located within the avoidance corridor of the simulated scenario experiment. This achievement underscores the potential of integrating kinematic and physiological data to deepen our understanding of how age impacts pedestrian avoidance velocity in pre-crash scenarios. Our study represents an advancement in the field of pedestrian safety research through applying a scaling approach to analyze active pedestrian responses. More importantly, this method can be further expanded to obtain the avoidance velocity of pedestrians, such as elderly individuals and women, who are rarely documented in stimulus experiments.

Engineering applications of the potential collision area

Active safety systems, such as Autonomous Emergency Braking (Edwards et al. 2014), commonly rely on predefined potential collision areas to assess collision risks (Roth et al.

2016; Ka et al. 2019; Dafrallah et al. 2021). Usually, these systems assume pedestrians walk at a uniform velocity and interact with vehicles. However, frequent braking in crowded urban areas may influence traffic efficiency, increase the risk of rear-end collisions and negatively affect the occupants' driving experience. In addition, ignoring pedestrian avoidance behavior will result in inaccuracies in triggering proactive safety features, such as the activation of pop-up engine hoods. This oversight can diminish the effectiveness of passive safety elements like bumpers (Smith et al. 2002) and pedestrian airbags (Lim et al. 2015). Our research, as outlined in Liu et al. (2023), suggests that incorporating pedestrian avoidance behaviors into the calculation of potential collision areas can significantly refine the precision of advanced vehicle protection systems. This offers precise input for advanced vehicle protection systems and reduces unnecessary interventions, especially in urban traffic environments.

The proposed potential collision areas have two distinctive features. First, the shape of these area is significantly influenced by pedestrians' decisions such as normal walking or avoidance. Second, there is a noticeable age-related trend: the size of these areas tends to increase with pedestrian's age. Based on these observations, we propose three critical guidelines for implementing active safety protection systems. Firstly, active safety protection systems should incorporate early warnings to pedestrians, allowing them to avoid approaching vehicles actively. Secondly, potential collision areas can serve as a basis for assessing the probability of collisions with pedestrians. In cases where the protection system determines that a pedestrian can safely avoid the crash, the vehicle can moderately reduce its velocity to ensure driving comfort and efficiency. Conversely, if a collision is unavoidable, the protection system should pre-optimize the passive structures (e.g., bumpers, airbags) to reduce the injury severity of pedestrians. Finally, with advancements in perception technology, intelligent connected systems, and cloud control systems (Chu et al. 2021), we anticipate that future HAVs will be able to differentiate among various age groups of pedestrians (Abdullah et al. 2021; Song et al. 2021). For groups such as the elderly, females, and children, HAVs should actively adjust their responses and adopt more cautious protection strategies due to their declining ability to avoid collisions and increased vulnerability to injuries. For young adults, who exhibit high athletic ability, HAVs can appropriately implement light braking when the collision risk is low. The above protective strategy ensures effective protection for vulnerable groups and minimizes the issues caused by overprotection, such as frequent braking, rear-end collisions, and traffic congestion. This is crucial in complex environments with heavy pedestrian traffic and busy roads, where HAVs must balance safety and traffic efficiency.

Limitations and future applications

Several limitations of the present study need to be considered. First, it is worth considering enhancing the realism of the experiment scenario to facilitate a more natural imitation

of avoidance behaviors by the subjects. This helps reduce the fatigue level of subjects in repeated trials, ensuring the validity of the data. Additionally, expanding the sample size and including a wider range of demographic groups, such as female adults and children, would improve the representativeness of the simulated scenario experiment. Meanwhile, the generalizability of the findings from the experiment, which focused on a standard crossroads pre-crash scenario, to other traffic scenarios remains to be validated. Additionally, the scaling approach only accounts for age factors, ignoring other pedestrians who may be at higher risk in traffic accidents, such as children and disabled people. Finally, while the potential collision areas focus on pedestrians' kinematic responses, it is also crucial to explore the impact of the vehicle dynamics. To overcome the mentioned limitations, future studies should broaden the range and fidelity of the subjects and scenarios while also including pedestrians' and vehicles' kinematic characteristics to enhance the accuracy of potential collision areas.

Conclusions

In this study, we developed and conducted a simulated scenario experiment and a scaling approach to determine the avoidance velocity for pedestrians of different ages. This methodology effectively addresses the previously encountered challenges of age-related limitations and the non-repeatability inherent in simulated scenario experiments. We also proposed a method for identifying potential collision areas for active safety systems that consider both pedestrians' age and avoidance velocity. Our findings reveal a clear trend: as pedestrians age, their capacity to avoid hazards diminishes, leading to a proportional increase in the size of potential collision areas. By analyzing the kinematic velocity of pedestrians across different ages, the resulting method enhances the precision in identifying potential collision areas in typical pre-crash scenarios. The proposed areas can serve as a valuable reference for designing active safety protection systems for highly automated vehicles to minimize pedestrians' collision risks in urban traffic environments.

Disclosure statement

We have not any relevant financial or non-financial competing interests.

Funding

This study is supported by the National Natural Science Foundation of China (52072216, 52131201). We would like to extend our sincere gratitude to the Tsinghua University-DiDi Joint Research Center for Future Mobility for funding pedestrian safety research.

ORCID

Siyuan Liu  <http://orcid.org/0009-0005-6751-3542>
 Qing Zhou  <http://orcid.org/0000-0002-5352-5842>
 Bingbing Nie  <http://orcid.org/0000-0002-8529-8613>

Data availability statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

References

- Abdullah NY, Ghazal MT, Waisi N. 2021. Pedestrian age estimation based on deep learning. *IJECS*. 22(3):1548–1555. doi:10.11591/ijeecs.v22.i3.pp1548-1555.
- Aoyagi Y, Shephard RJ. 1992. Aging and muscle function. *Sports Med*. 14(6):376–396. doi:10.2165/00007256-199214060-00005.
- Chu W, Wuniri Q, Du X, Xiong Q, Huang T, Li K. 2021. Cloud control system architectures, technologies and applications on intelligent and connected vehicles: a review. *Chin J Mech Eng*. 34(1):139. doi:10.1186/s10033-021-00638-4.
- Coelingh E, Eidehall A, Bengtsson M. 2010. Collision warning with full auto brake and pedestrian detection—a practical example of automatic emergency braking. In 13th International IEEE Conference on Intelligent Transportation Systems (pp. 155–160). IEEE.
- Dafraallah S, Amine A, Mousset S, Benshrar A. 2021. Monocular pedestrian orientation recognition based on capsule network for a novel collision warning system. *IEEE*. 9:141635–141650. doi:10.1109/ACCESS.2021.3119629.
- Der G, Deary IJ. 2006. Age and sex differences in reaction time in adulthood: results from the United Kingdom Health and Lifestyle Survey. *Psychol Aging*. 21(1):62–73. doi:10.1037/0882-7974.21.1.62.
- Edwards M, Nathanson A, Carroll J, Wisch M, Zander O, Lubbe N. 2015. Assessment of integrated pedestrian protection systems with autonomous emergency braking (AEB) and passive safety components. *Traffic Inj Prev*. 16 (sup1):S2–S11. doi:10.1080/15389588.2014.1003154.
- Edwards M, Nathanson A, Wisch M. 2014. Estimate of potential benefit for Europe of fitting Autonomous Emergency Braking (AEB) systems for pedestrian protection to passenger cars. *Traffic Inj Prev*. 15(sup1):S173–S82. doi:10.1080/15389588.2014.931579.
- Gerogiannis A, Bode NW. 2024. Analysis of long-term observational data on pedestrian road crossings at unmarked locations. *Saf Sci*. 172:106420. doi:10.1016/j.ssci.2024.106420.
- Grimmer M, Riener R, Walsh CJ, Seyfarth A. 2019. Mobility related physical and functional losses due to aging and disease – a motivation for lower limb exoskeletons. *J Neuroeng Rehabil*. 16(1):2. doi:10.1186/s12984-018-0458-8.
- Gunji Y, Aoki T, Okamura K, Ito Y. 2019. Investigation of restraint characteristics for elderly occupant chest injury reduction. In 26th International Technical Conference on the Enhanced Safety of Vehicles (ESV): Technology: Enabling a Safer Tomorrow National Highway Traffic Safety Administration (No. 19-0275).
- Harris GF, Wertsch JJ. 1994. Procedures for gait analysis. *Arch Phys Med Rehabil*. 75(2):216–225. doi:10.1016/0003-9993(94)90399-9.
- Iosa M, Fusco A, Marchetti F, Morone G, Caltagirone C, Paolucci S, Peppe A. 2013. The golden ratio of gait harmony: repetitive proportions of repetitive gait phases. *Biomed Res Int*. 2013:918642–918647. doi:10.1155/2013/918642.
- Ka D, Lee D, Kim S, Yeo H. 2019. Study on the framework of intersection pedestrian collision warning system considering pedestrian characteristics. *Transportation Research Record*. 2673(5):747–758. doi:10.1177/0361198119838519.
- Kerrigan DC, Todd MK, Della Croce U. 1998. Gender differences in joint biomechanics during walking: normative study in young adults. *Am J Phys Med Rehabil*. 77(1):2–7. doi:10.1097/00002060-199801000-00002.
- Kim JK, Ulfarsson GF, Shankar VN, Kim S. 2008. Age and pedestrian injury severity in motor-vehicle crashes: a heteroskedastic logit analysis. *Accid Anal Prev*. 40(5):1695–1702. doi:10.1016/j.aap.2008.06.005.
- Knoblauch RL, Pietrucha MT, Nitzburg M. 1996. Field studies of pedestrian walking speed and start-up time. *Transportation Research Record*. 1538(1):27–38. doi:10.1177/0361198196153800104.
- Li Q, Gan S, Shang S, Cui TS, Zhou Q, Nie B. 2021. Whole-Body Motion Category in Pedestrian-Vehicle Crashes Identified via Volunteer Tests and Publicly Available Videos. In 2021 IRCOBI Asia

- conference proceedings. Beijing, China: International Research Council on the Biomechanics of Injury (IRCOBI). p. 52–55.
- Lim JH, Park JS, Yun YW, Jeong SB, Park GJ. 2015. Design of an air-bag system of a mid-sized automobile for pedestrian protection. *Proceedings of the Institution of Mechanical Engineers, Part D: Journal of Automobile Engineering*. 229(5):656–669. doi:[10.1177/0954407014551186](https://doi.org/10.1177/0954407014551186).
- Li Q, Shang S, Pei X, Wang Q, Zhou Q, Nie B. 2021. Kinetic and kinematic features of pedestrian avoidance behavior in motor vehicle conflicts. *Front Bioeng Biotechnol*. 9:783003. doi:[10.3389/fbioe.2021.783003](https://doi.org/10.3389/fbioe.2021.783003).
- Liu S, Li Q, Zhou Q, Nie B. 2023. Constructing a virtual testing platform on pedestrian safety incorporating active human behavior for highly automated vehicles. In 2023 IRCOBI conference proceedings. Cambridge, UK: International Research Council on the Biomechanics of Injury (IRCOBI). p. 529–530.
- Liu YC, Tung YC. 2014. Risk analysis of pedestrians' road-crossing decisions: effects of age, time gap, time of day, and vehicle speed. *Saf Sci*. 63:77–82. doi:[10.1016/j.ssci.2013.11.002](https://doi.org/10.1016/j.ssci.2013.11.002).
- Nie B, Li Q, Gan S, Xing B, Huang Y, Li SE. 2021. Safety envelope of pedestrians upon motor vehicle conflicts identified via active avoidance behaviour. *Sci Rep*. 11(1):3996. doi:[10.1038/s41598-021-82331-z](https://doi.org/10.1038/s41598-021-82331-z).
- Niebuhr T, Junge M, Rosén E. 2016. Pedestrian injury risk and the effect of age. *Accid Anal Prev*. 86:121–128. doi:[10.1016/j.aap.2015.10.026](https://doi.org/10.1016/j.aap.2015.10.026).
- Oikawa S, Matsui Y, Doi T, Sakurai T. 2016. Relation between vehicle travel velocity and pedestrian injury risk in different age groups for the design of a pedestrian detection system. *Saf Sci*. 82:361–367. doi:[10.1016/j.ssci.2015.10.003](https://doi.org/10.1016/j.ssci.2015.10.003).
- Park MK, Lee SY, Kwon CK, Kim SW. 2016. Design of pedestrian target selection with funnel map for pedestrian AEB system. *IEEE Trans Veh Technol*. 66(5):1–1. doi:[10.1109/TVT.2016.2604420](https://doi.org/10.1109/TVT.2016.2604420).
- Pasco JA, Stuart AL, Holloway-Kew KL, Tembo MC, Sui SX, Anderson KB, Hyde NK, Williams LJ, Kotowicz MA. 2020. Lower-limb muscle strength: normative data from an observational population-based study. *BMC Musculoskelet Disord*. 21(1):89. doi:[10.1186/s12891-020-3098-7](https://doi.org/10.1186/s12891-020-3098-7).
- Roth M, Flohr F, Gavril DM. 2016. June. Driver and pedestrian awareness-based collision risk analysis. In 2016 IEEE Intelligent Vehicles Symposium (IV) (pp. 454–459). IEEE. doi:[10.1109/IVS.2016.7535425](https://doi.org/10.1109/IVS.2016.7535425).
- Schachner M, Schneider B, Klug C, Sinz W. 2020. Extracting Quantitative Descriptions of Pedestrian Pre-crash Postures from Real-World AccidentVideos. In 2020 IRCOBI conference proceedings. Farmington Hills (MI): International Research Council on the Biomechanics of Injury (IRCOBI). p. 231–249.
- Smith A, Ashmead M, Gay P, Siever N. 2002. Pedestrian protection potential of an adaptive bumper. Huntingdon (UK): Cellbond Composites Ltd.
- Song T, Yang X, Wang Y, Lei Y, Liu G. 2021. Pedestrian age recognition method based on gait deep learning. *J Phys: conf Ser*. 2010(1):012031. doi:[10.1088/1742-6596/2010/1/012031](https://doi.org/10.1088/1742-6596/2010/1/012031).
- Soni A, Robert T, Rongieras F, Beillas P. 2013. Observations on pedestrian pre-crash reactions during simulated accidents. *Stapp Car Crash J*. 57:157–183. doi:[10.4271/2013-22-0006](https://doi.org/10.4271/2013-22-0006).
- Soni A, Robert T, Beillas P. 2013. Effects of Pedestrian Pre-Crash Reactions on Crash Outcomes during Multi-body Simulations. In IRCOBI Conference Proceedings (No. IRC-13-92).
- Tian K, Markkula G, Wei C, Lee YM, Madigan R, Merat N, Romano R. 2022. Explaining unsafe pedestrian road crossing behaviours using a psychophysics-based gap acceptance model. *Saf Sci*. 154:105837. doi:[10.1016/j.ssci.2022.105837](https://doi.org/10.1016/j.ssci.2022.105837).
- World Health Organization. 2023. Global status report on road safety 2023. Geneva, Switzerland: World Health Organization.
- Ye C, Wang X, Morris A, Ying Z. 2024. Pedestrian crash causation analysis and active safety system calibration. *Accid Anal Prev*. 195:107404. doi:[10.1016/j.aap.2023.107404](https://doi.org/10.1016/j.aap.2023.107404).
- Zhou Q, Rouhana SW, Melvin JW. 1996. Age effects on thoracic injury tolerance. *SAE Trans*. 105:1819–1830.
- Zou T, Zha A, Liu Q, Simms C. 2022. Pedestrian gaits observed from actual pedestrian-vehicle collisions. *Int J Crashworthiness*. 27(1):1–23. doi:[10.1080/13588265.2020.1769455](https://doi.org/10.1080/13588265.2020.1769455).